

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Product Description:	Methyl vinyl ketone, stabilized
Cat No. :	C12800
Synonyms	3-Buten-2-one; MVK
CAS No	78-94-4
Molecular Formula	C ₄ H ₆ O

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Uses advised against	No Information available

Details of the supplier of the safety data sheet

Importer	Supplier
Fisher Scientific Korea	Thermo Fisher Scientific Chemicals, Inc.
D5,D6, Incheon Airport Logistics Complex	30 Bond Street
150, Gonghangdong-Ro 296 Beon-Gil	Ward Hill, MA 01835-8099
Jung-Gu, Incheon	
Tel: +82-1661-9555	
Fax: +82-2-2023-0603	

E-mail address	Chem.KR@thermofisher.com
-----------------------	--------------------------

Emergency Telephone Number

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970
 CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture
Physical hazards

Flammable liquids	Category 2
-------------------	------------

Health hazards

Acute Oral Toxicity	Category 2
Acute Dermal Toxicity	Category 1
Acute Inhalation Toxicity - Vapors	Category 1
Skin Corrosion/Irritation	Category 1 B
Serious Eye Damage/Eye Irritation	Category 1
Skin Sensitization	Category 1
Specific target organ toxicity - (repeated exposure)	Category 2

Environmental hazards

SAFETY DATA SHEET

Methyl vinyl ketone, stabilized

Revision Date 12-Jun-2024

Acute aquatic toxicity
Chronic aquatic toxicity

Category 1
Category 1

Label Elements



Signal Word

Danger

Hazard Statements

H373 - May cause damage to organs through prolonged or repeated exposure
H410 - Very toxic to aquatic life with long lasting effects
H225 - Highly flammable liquid and vapor
H314 - Causes severe skin burns and eye damage
H317 - May cause an allergic skin reaction
H400 - Very toxic to aquatic life
H318 - Causes serious eye damage
H300 + H310 + H330 - Fatal if swallowed, in contact with skin or if inhaled

Precautionary Statements

Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
P240 - Ground and bond container and receiving equipment
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment
P242 - Use non-sparking tools
P243 - Take action to prevent static discharges
P262 - Do not get in eyes, on skin, or on clothing
P264 - Wash face, hands and any exposed skin thoroughly after handling
P270 - Do not eat, drink or smoke when using this product
P271 - Use only outdoors or in a well-ventilated area
P272 - Contaminated work clothing should not be allowed out of the workplace
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P284 - Wear respiratory protection

Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor
P330 - Rinse mouth
P331 - Do NOT induce vomiting
P363 - Wash contaminated clothing before reuse
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction

Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed
P405 - Store locked up

Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

Other Hazards

This product does not contain any known or suspected endocrine disruptors

SAFETY DATA SHEET

Methyl vinyl ketone, stabilized

Revision Date 12-Jun-2024

NFPA

Health
4

Flammability
3

Instability
1

Physical hazards
N/A

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	Common Name	CAS No	Index No	Weight %
Methyl vinyl ketone	3-Buten-2-one; MVK	78-94-4	KE-04112	95 - 97.5
Water	Aqua	7732-18-5	KE-35400	2.5 - <5
Acetic acid	Acetic acid	64-19-7	KE-00013	0.5 - 1
Acetonitrile	AN; Methyl cyanide; Ethanenitrile	75-05-8	KE-00067	0.5 - 1
Hydroquinone	1,4-Dihydroxybenzene; 1,4-Benzenediol	123-31-9	KE-35112	0.1 - 0.5

SECTION 4: FIRST AID MEASURES

Description of first aid measures

General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

Inhalation

If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.

Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. May cause allergic skin reaction. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

SAFETY DATA SHEET

Methyl vinyl ketone, stabilized

Revision Date 12-Jun-2024

Extinguishing media

Suitable Extinguishing Media

Water mist may be used to cool closed containers. CO₂, dry chemical, dry sand, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

No information available.

Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Do not allow run-off from fire-fighting to enter drains or water courses.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂), Thermal decomposition can lead to release of irritating gases and vapors.

Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Keep away from heat, sparks and flame. To maintain product quality: Keep refrigerated. Keep at temperatures below 10°C.

Specific End Uses

Use in laboratories.

SAFETY DATA SHEET

Methyl vinyl ketone, stabilized

Revision Date 12-Jun-2024

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Component	CAS No	Korea	ACGIH TLV	OSHA PEL
Methyl vinyl ketone	78-94-4	Not listed	Ceiling: 0.01 ppm	Not listed
Water	7732-18-5	Not listed	Not listed	Not listed
Acetic acid	64-19-7	STEL: 15 ppm TWA: 10 ppm	TWA: 10 ppm STEL: 15 ppm	(Vacated) TWA: 10 ppm (Vacated) TWA: 25 mg/m ³ TWA: 10 ppm TWA: 25 mg/m ³
Acetonitrile	75-05-8	TWA: 20 ppm TWA: 5 mg/m ³ Skin	TWA: 20 ppm Skin	(Vacated) TWA: 40 ppm (Vacated) TWA: 70 mg/m ³ (Vacated) TWA: 5 mg/m ³ (Vacated) STEL: 60 ppm (Vacated) STEL: 105 mg/m ³ TWA: 40 ppm TWA: 70 mg/m ³
Hydroquinone	123-31-9	TWA: 2 mg/m ³	TWA: 1 mg/m ³	(Vacated) TWA: 2 mg/m ³ TWA: 2 mg/m ³

Component	CAS No	European Union	The United Kingdom	Germany
Methyl vinyl ketone	78-94-4	Not listed	Not listed	Haut
Water	7732-18-5	Not listed	Not listed	Not listed
Acetic acid	64-19-7	TWA: 25 mg/m ³ (8h) TWA: 10 ppm (8h) STEL: 50 mg/m ³ (15min) STEL: 20 ppm (15min)	STEL: 37 mg/m ³ STEL: 15 ppm TWA: 10 ppm TWA: 25 mg/m ³	TWA: 10 ppm (8 Stunden). AGW - exposure factor 2 TWA: 25 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 10 ppm (8 Stunden). MAK TWA: 25 mg/m ³ (8 Stunden). MAK Höhepunkt: 20 ppm Höhepunkt: 50 mg/m ³
Acetonitrile	75-05-8	TWA: 40 ppm (8hr) TWA: 70 mg/m ³ (8hr) Skin	STEL: 60 ppm 15 min STEL: 102 mg/m ³ 15 min TWA: 40 ppm 8 hr TWA: 68 mg/m ³ 8 hr	TWA: 10 ppm (8 Stunden). AGW - exposure factor 2 TWA: 17 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 10 ppm (8 Stunden). MAK TWA: 17 mg/m ³ (8 Stunden). MAK TWA: 2 mg/m ³ (8 Stunden). MAK Höhepunkt: 20 ppm Höhepunkt: 34 mg/m ³ Höhepunkt: 2 mg/m ³ Haut
Hydroquinone	123-31-9	Not listed	STEL: 1.5 mg/m ³ 15 min TWA: 0.5 mg/m ³ 8 hr	Haut

ACGIH - Biological Exposure Indices

Component	CAS No	ACGIH - Biological Exposure Indices
Methyl vinyl ketone	78-94-4	Not listed
Water	7732-18-5	Not listed
Acetic acid	64-19-7	Not listed
Acetonitrile	75-05-8	Not listed
Hydroquinone	123-31-9	Not listed

Exposure Controls

Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to

SAFETY DATA SHEET

Methyl vinyl ketone, stabilized

Revision Date 12-Jun-2024

control hazardous materials at source

Personal protective equipment

Eye Protection	Goggles
Hand Protection	Protective gloves
Skin and body protection	Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Personal protective equipment	Use only those certified by the Korea Occupational Safety and Health Administration.
Respiratory Protection	When workers are facing concentrations above the exposure limit they must use appropriate certified respirators
Recommended Filter type:	Organic gases and vapours filter Type A Brown conforming to EN14387
	To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly
	When RPE is used a face piece Fit Test should be conducted

<u>Hygiene Measures</u>	Handle in accordance with good industrial hygiene and safety practice
--------------------------------	---

<u>Environmental exposure controls</u>	Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained
---	--

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance (Physical State, Color, etc.) Light yellow Liquid

Odor	Irritating pungent
Odor Threshold	0.2 ppm
pH	Not applicable

Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	80 °C / 176 °F	@ 760 mmHg
Flash Point	-7 °C / 19.4 °F	Method - No information available

Evaporation Rate	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	Lower 2.1 vol% Upper 15.6 vol%	

Vapor Pressure	130 mbar @ 28 °C	
Vapor Density	2.4	(Air = 1.0)
Specific Gravity / Density	0.864 g/cm3 @20°C	
Bulk Density	Not applicable	Liquid
Water Solubility	Miscible	
Solubility in other solvents	No information available	

SAFETY DATA SHEET

Methyl vinyl ketone, stabilized

Revision Date 12-Jun-2024

Partition Coefficient (n-octanol/water)

Component	CAS No	log Pow
Methyl vinyl ketone	78-94-4	No data available
Water	7732-18-5	No data available
Acetic acid	64-19-7	-0.2
Acetonitrile	75-05-8	-0.34
Hydroquinone	123-31-9	0.59

Autoignition Temperature	370 °C / 698 °F	
Decomposition Temperature	No data available	
Viscosity	Dynamic 0.81 mPa.s (70 °C)	
Explosive Properties		Vapors may form explosive mixtures with air
Oxidizing Properties	No information available	
Molecular Formula	C4 H6 O	
Molecular Weight	70.09	
Surface tension	24 mN/m	

SECTION 10: STABILITY AND REACTIVITY

Reactivity Yes Hazardous polymerization may occur

Chemical Stability Light sensitive. heat sensitive.

Possibility of Hazardous Reactions

Hazardous Polymerization Polymerization can occur.
Hazardous Reactions None under normal processing.

Conditions to Avoid Excess heat. Exposure to light. Keep away from open flames, hot surfaces and sources of ignition. Incompatible products.

Incompatible Materials Oxidizing agent. Reducing Agent. oxygen. Bases. Amines. Ammonia.

Hazardous Decomposition Products Carbon monoxide (CO). Carbon dioxide (CO₂). Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

Information on expected route of exposure
Inhalation May produce an allergic reaction.

SAFETY DATA SHEET

Methyl vinyl ketone, stabilized

Revision Date 12-Jun-2024

Ingestion May cause allergic reaction. May be harmful if swallowed.
Eyes Avoid contact with eyes. Corrosive to the eyes and may cause severe damage including blindness. Sensitization.
Skin Avoid contact with skin. Causes burns. Skin Corrosion/Irritation. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons. Harmful in contact with skin.

Information on Health Hazards

(a) acute toxicity;

Oral Category 2
Dermal Category 1
Inhalation Category 1

Toxicology data for the components

Component	CAS No	LD50 Oral	LD50 Dermal	LC50 Inhalation
Methyl vinyl ketone	78-94-4	LD50 = 23.1 mg/kg (Rat)	LD50 = 35 mg/kg (Rat)	LC50 = 7 mg/m ³ (Rat) 4 h
Water	7732-18-5	-	-	-
Acetic acid	64-19-7	3310 mg/kg (Rat)	-	> 40 mg/L (Rat) 4 h
Acetonitrile	75-05-8	450-787 mg/kg (Rat) 2460 mg/kg (Rat)	> 2000 mg/kg (Rabbit)	LC50 = 3587 ppm (6.022 mg/l) (Mouse) 4h LC50 = 16,000 ppm (26.8 mg/l) (Rat) 4h
Hydroquinone	123-31-9	LD50 = 298 mg/kg (Rat)	LD50 = 74800 mg/kg (Rabbit)	No data available

(b) skin corrosion/irritation; Category 1

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory No data available
Skin Category 1

Component	CAS No	Test method	Test species	Study result
Methyl vinyl ketone	78-94-4	No data available	No data available	No data available
Water	7732-18-5	No data available	No data available	No data available
Acetic acid	64-19-7	No data available	No data available	No data available
Acetonitrile	75-05-8	No data available	No data available	No data available
Hydroquinone	123-31-9	No data available	No data available	No data available

May cause sensitization by skin contact

(e) germ cell mutagenicity; No data available

Component	CAS No	Test method	Test species	Study result
Methyl vinyl ketone	78-94-4	No data available	No data available	No data available
Water	7732-18-5	No data available	No data available	No data available
Acetic acid	64-19-7	No data available	No data available	No data available
Acetonitrile	75-05-8	No data available	No data available	No data available
Hydroquinone	123-31-9	No data available	No data available	No data available

(f) carcinogenicity; No data available

Component	CAS No	Test method	Test species / Duration	Study result
Methyl vinyl ketone	78-94-4	No data available	No data available	No data available
Water	7732-18-5	No data available	No data available	No data available
Acetic acid	64-19-7	No data available	No data available	No data available

SAFETY DATA SHEET

Methyl vinyl ketone, stabilized

Revision Date 12-Jun-2024

Acetonitrile	75-05-8	No data available	No data available	No data available
Hydroquinone	123-31-9	No data available	No data available	No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

Component	CAS No	IARC	NTP	ACGIH	OSHA	UK
Methyl vinyl ketone	78-94-4	Not listed	Not listed	Not listed	Not listed	Not listed
Water	7732-18-5	Not listed	Not listed	Not listed	Not listed	Not listed
Acetic acid	64-19-7	Not listed	Not listed	Not listed	Not listed	Not listed
Acetonitrile	75-05-8	Not listed	Not listed	Not listed	Not listed	Not listed
Hydroquinone	123-31-9	Not listed	Not listed	A3	Not listed	Not listed

ACGIH: (American Conference of Governmental Industrial Hygienists)
 A1 - Known Human Carcinogen
 A2 - Suspected Human Carcinogen
 A3 - Animal Carcinogen
 ACGIH: (American Conference of Governmental Industrial Hygienists)

(g) reproductive toxicity; No data available

Component	CAS No	Test method	Test species / Duration	Study result
Methyl vinyl ketone	78-94-4	No data available	No data available	No data available
Water	7732-18-5	No data available	No data available	No data available
Acetic acid	64-19-7	No data available	No data available	No data available
Acetonitrile	75-05-8	No data available	No data available	No data available
Hydroquinone	123-31-9	No data available	No data available	No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; Category 2

Target Organs Kidney, Central nervous system (CNS), Lungs.

(j) aspiration hazard; No data available

Other Adverse Effects

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

Component	CAS No	EU - Endocrine Disruptors Candidate List	EU - Endocrine Disruptors - Evaluated Substances	Japan - Endocrine Disruptor Information
Methyl vinyl ketone	78-94-4	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Acetic acid	64-19-7	Not applicable	Not applicable	Not applicable
Acetonitrile	75-05-8	Not applicable	Not applicable	Not applicable
Hydroquinone	123-31-9	Not applicable	Not applicable	Not applicable

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects

The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

Component	CAS No	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
Methyl vinyl ketone	78-94-4	No data available	No data available	No data available	No data available
Water	7732-18-5	No data available	No data available	No data available	No data available

ALFAAC12800

SAFETY DATA SHEET

Methyl vinyl ketone, stabilized

Revision Date 12-Jun-2024

Acetic acid	64-19-7	Pimephales promelas: LC50 = 88 mg/L/96h Lepomis macrochirus: LC50 = 75 mg/L/96h	EC50 = 95 mg/L/24h	-	Photobacterium phosphoreum: EC50 = 8.8 mg/L/15 min Photobacterium phosphoreum: EC50 = 8.8 mg/L/25 min Photobacterium phosphoreum: EC50 = 8.8 mg/L/5 min
Acetonitrile	75-05-8	LC50: = 1850 mg/L, 96h static (Lepomis macrochirus) LC50: = 1000 mg/L, 96h static (Pimephales promelas) LC50: 1600 - 1690 mg/L, 96h flow-through (Pimephales promelas) LC50: = 1650 mg/L, 96h static (Poecilia reticulata)	No data available	No data available	EC50 = 28000 mg/L 48 h EC50 = 73 mg/L 24 h EC50 = 7500 mg/L 15 h
Hydroquinone	123-31-9	LC50: 0.1 - 0.18 mg/L, 96h static (Pimephales promelas) LC50: = 0.17 mg/L, 96h (Brachydanio rerio) LC50: = 0.044 mg/L, 96h flow-through (Pimephales promelas) LC50: = 0.044 mg/L, 96h flow-through (Oncorhynchus mykiss)	EC50: = 0.29 mg/L, 48h (Daphnia magna)	EC50: = 0.335 mg/L, 72h (Pseudokirchneriella subcapitata)	EC50 = 0.038 mg/L 15 min EC50 = 0.0382 mg/L 30 min EC50 = 0.042 mg/L 5 min EC50 = 23.75 mg/L 60 min

Persistence and degradability

Persistence

Degradation in sewage treatment plant

Not readily biodegradable

Persistence is unlikely, based on information available.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Acetic acid	-0.2	No data available
Acetonitrile	-0.34	No data available
Hydroquinone	0.59	40 dimensionless

Mobility in soil

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

Surface tension

24 mN/m

Ozone Depletion Potential

Component	CAS No	Ozone Depletion Potential
Methyl vinyl ketone	78-94-4	Not listed
Water	7732-18-5	Not listed
Acetic acid	64-19-7	Not listed
Acetonitrile	75-05-8	Not listed

SAFETY DATA SHEET

Methyl vinyl ketone, stabilized

Revision Date 12-Jun-2024

Hydroquinone	123-31-9	Not listed
--------------	----------	------------

Other adverse effects No information available

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste from Residues/Unused Products	Waste is classified as hazardous. Dispose in accordance with the Wastes Control Act (폐기물관리법).
Contaminated Packaging	Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.
Other Information	Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment.

SECTION 14: TRANSPORT INFORMATION

Road and Rail Transport

UN-No	UN1251
Proper Shipping Name	METHYL VINYL KETONE, STABILIZED
Hazard Class	6.1
Subsidiary Hazard Class	3, 8
Packing Group	I

IATA

UN-No	UN1251
Proper Shipping Name	METHYL VINYL KETONE, STABILIZED FORBIDDEN FOR IATA TRANSPORT
Hazard Class	6.1
Subsidiary Hazard Class	3, 8
Packing Group	I

IMDG/IMO

UN-No	UN1251
Proper Shipping Name	METHYL VINYL KETONE, STABILIZED
Hazard Class	6.1
Subsidiary Hazard Class	3, 8
Packing Group	I
Marine Pollutant	Dangerous for the environment Product is a marine pollutant according to the criteria set by IMDG/IMO

Special Precautions for User	Storage conditions in Section 7 should also be met during transportation Cooled transportation <10°C is recommended to ensure shelf-life Inhibitors have been added to stabilize this product Inhibitor levels should be maintained Hazardous polymerization may occur upon depletion of inhibitor
-------------------------------------	--

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

International Inventories

ALFAAC12800

SAFETY DATA SHEET

Methyl vinyl ketone, stabilized

Revision Date 12-Jun-2024

Component	CAS No	KECL	TSCA	EINECS	IECSC	DSL	NDSL	PICCS	ENCS	ISHL	AICS
Methyl vinyl ketone	78-94-4	KE-04112	X	201-160-6	X	-	X	X	X	X	X
Water	7732-18-5	KE-35400	X	231-791-2	X	X	-	X	X		X
Acetic acid	64-19-7	X	X	200-580-7	X	X	-	X	X	X	X
Acetonitrile	75-05-8	KE-00067	X	200-835-2	X	X	-	X	X	X	X
Hydroquinone	123-31-9	KE-35112	X	204-617-8	X	X	-	X	X	X	X

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Methyl vinyl ketone	78-94-4	Not applicable	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable
Acetic acid	64-19-7	Not applicable	Not applicable	Not applicable	Annex I - Y34
Acetonitrile	75-05-8	Not applicable	Not applicable	Not applicable	Not applicable
Hydroquinone	123-31-9	Not applicable	Not applicable	Not applicable	Annex I - Y39

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
Methyl vinyl ketone	78-94-4	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Listed	Not applicable	Not applicable
Acetic acid	64-19-7	Listed	Not applicable	Not applicable
Acetonitrile	75-05-8	Listed	Not applicable	Not applicable
Hydroquinone	123-31-9	Listed	Not applicable	Not applicable

Korean National Regulations

Component	CAS No	Act on Registration and Evaluation of Chemical Substances (K-REACH)	Authorised Chemicals	Existing Substances Subject to Registration
Methyl vinyl ketone	78-94-4	Annex 1 - KE-04112	Not applicable	Not applicable
Water	7732-18-5	Annex 1 - KE-35400 Exempt (Index No. 25)	Not applicable	Not applicable
Acetic acid	64-19-7	Annex 1 - KE-00013	Not applicable	Not applicable
Acetonitrile	75-05-8	Annex 1 - KE-00067	Not applicable	Listed
Hydroquinone	123-31-9	Annex 1 - KE-35112	Not applicable	Listed

Component	CAS No	Chemical Control Act - Toxic Chemicals	Chemical Control Act - Prohibited Chemicals	Chemical Control Act - Use Restricted Chemicals
Methyl vinyl ketone	78-94-4	1997-1-0078 (>1%)	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Acetic acid	64-19-7	2023-1-1147 (>=25%)	Not applicable	Not applicable
Acetonitrile	75-05-8	2021-1-1056 (>25%)	Not applicable	Not applicable
Hydroquinone	123-31-9	2010-1-0610 (>2.5%)	Not applicable	Not applicable

Component	CAS No	Chemical Control Act - Accident Precaution Chemicals (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures)
Methyl vinyl ketone	78-94-4	>1%	200000 kg/yr	1500000 kg/yr
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Acetic acid	64-19-7	Not applicable	Not applicable	Not applicable
Acetonitrile	75-05-8	Not applicable	Not applicable	Not applicable
Hydroquinone	123-31-9	Not applicable	Not applicable	Not applicable

Component	CAS No	Waste Control Law	Ministry of Environment - CMR risk	Ministry of Environment - Critically Controlled Substance

ALFAAC12800

SAFETY DATA SHEET

Methyl vinyl ketone, stabilized

Revision Date 12-Jun-2024

Methyl vinyl ketone	78-94-4	> 1% (CCA)	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Acetic acid	64-19-7	Not applicable	Not applicable	Not applicable
Acetonitrile	75-05-8	>1 mg/L (WCL) > 25% (CCA)	Not applicable	Not applicable
Hydroquinone	123-31-9	> 2.5% (CCA)	Not applicable	Not applicable

CCA = Chemical Control Act

WCL = Waste Control Law

Component	CAS No	ISHA - Harmful Agents Subject to Work Environment Monitoring	ISHA - Prohibited substances	ISHA - Substances requiring permission
Methyl vinyl ketone	78-94-4	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Acetic acid	64-19-7	Listed	Not applicable	Not applicable
Acetonitrile	75-05-8	Listed	Not applicable	Not applicable
Hydroquinone	123-31-9	Listed	Not applicable	Not applicable

Component	CAS No	ISHA - Substances subject to control	ISHA - Harmful Agents Requiring Health Examination	ISHA - Permissible Exposure Limits
Methyl vinyl ketone	78-94-4	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Acetic acid	64-19-7	Listed	Not applicable	Not applicable
Acetonitrile	75-05-8	Listed	Listed	Not applicable
Hydroquinone	123-31-9	Listed	Listed	Not applicable

Component	CAS No	ISHA - Subject to Process Safety Reports (minimum quantity)	ISHA - Threshold Limit Values (TLVs) Chemicals	ISHA - Special management materials
Methyl vinyl ketone	78-94-4	5000 kg	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Acetic acid	64-19-7	5000 kg	STEL: 15 ppm TWA: 10 ppm	Not applicable
Acetonitrile	75-05-8	5000 kg	TWA: 20 ppm TWA: 5 mg/m ³ Skin	Not applicable
Hydroquinone	123-31-9	Not applicable	TWA: 2 mg/m ³	Not applicable

National Fire Association - Dangerous Substances Minimum quantity requiring a permit

Component	CAS No	Class 1 - Oxidising solids	Class 2 - Flammable solid	Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet	Class 4 - Flammable liquids	Class 5 - Self-reactive substances	Class 6 - Oxidising liquids
Methyl vinyl ketone	78-94-4	Not applicable	Not applicable	Not applicable	2. Group 1 Petroleum (Insoluble) 200 L	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Acetic acid	64-19-7	Not applicable	Not applicable	Not applicable	4. Group 2 Petroleum (Soluble) 2000 L	Not applicable	Not applicable
Acetonitrile	75-05-8	Not applicable	Not applicable	Not applicable	2. Group 1 Petroleum (Soluble) 400 L	Not applicable	Not applicable
Hydroquinone	123-31-9	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable

Control Parameters

SAFETY DATA SHEET

Methyl vinyl ketone, stabilized

Revision Date 12-Jun-2024

Component	CAS No	Korea	ACGIH - Biological Exposure Indices
Methyl vinyl ketone	78-94-4	Not listed	Not listed
Water	7732-18-5	Not listed	Not listed
Acetic acid	64-19-7	STEL: 15 ppm TWA: 10 ppm	Not listed
Acetonitrile	75-05-8	TWA: 20 ppm TWA: 5 mg/m ³ Skin	Not listed
Hydroquinone	123-31-9	TWA: 2 mg/m ³	Not listed

US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

Component	CAS No	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Methyl vinyl ketone	78-94-4	Not applicable	TQ: 100 lb
Water	7732-18-5	Not applicable	Not applicable
Acetic acid	64-19-7	Not applicable	Not applicable
Acetonitrile	75-05-8	Not applicable	Not applicable
Hydroquinone	123-31-9	Not applicable	Not applicable

CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355)

Component	CAS No	CERCLA Extremely Hazardous Substances RQs	Hazardous Substances RQs	SARA 313 - Threshold Values %
Methyl vinyl ketone	78-94-4	10 lb	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Acetic acid	64-19-7	Not applicable	5000 lb	Not applicable
Acetonitrile	75-05-8	Not applicable	5000 lb	1.0 %
Hydroquinone	123-31-9	100 lb	100 lb	1.0 %

GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Danger.

H225 - Highly flammable liquid and vapor. H314 - Causes severe skin burns and eye damage. H317 - May cause an allergic skin reaction. H373 - May cause damage to organs through prolonged or repeated exposure. H410 - Very toxic to aquatic life with long lasting effects. H300 + H310 + H330 - Fatal if swallowed, in contact with skin or if inhaled.

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. P280 - Wear protective gloves/protective clothing/eye protection/face protection. P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower. P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing. P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P310 - Immediately call a POISON CENTER or doctor/physician.

SECTION 16: OTHER INFORMATION

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

SAFETY DATA SHEET

Methyl vinyl ketone, stabilized

Revision Date 12-Jun-2024

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC - (Volatile Organic Compound)

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.
First aid for chemical exposure, including the use of eye wash and safety showers.
Chemical incident response training.

Prepared By	Health, Safety and Environmental Department
Creation Date	07-Oct-2010
Revision Date	12-Jun-2024
Revision Number	3
Revision Summary	New emergency telephone response service provider.

MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet